

# The Integrated Framework: Stratified Realization, Information Integration, and EM Model Class Auditing

Vinay K. Awasthi  
Johns Hopkins University  
Department of Mathematics

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## Abstract

We establish a mathematically rigorous, publication-grade synthesis unifying algebraic operation networks (Layer 1), continuous information geometry and Baudot–Bennequin–Vigneaux (BBV) information cohomology (Layer 2), and discrete Abramsky–Mansfield–Barbosa (AMB) sheaf contextuality (Layer 3). Rather than asserting a naive cochain-level isomorphism across distinct mathematical sites, we formulate a stratified dynamical system governed by forward pushforward sampling maps and backward empirical pullbacks. We resolve cochain degree indexing ambiguities by explicitly distinguishing 0-coboundaries  $\delta^0 : C^0 \rightarrow C^1$  (vertex potentials) from 1-coboundaries  $\delta^1 : C^1 \rightarrow C^2$  (loop holonomies). Furthermore, we prove the preservation/erasure dichotomy under Expectation-Maximization (EM) optimization over latent model classes and establish the exponential sample complexity required for operational holonomy recovery under continuous noise.

## 1 Introduction and Stratified System Architecture

In accordance with the stratification principle, the interaction between operation networks, probability manifolds, and outcome sheaves is mediated through a dynamic feedback control loop rather than a direct cochain-level isomorphism:

$$\begin{array}{lcl}
 \textbf{Layer 1 (Operations)} & H^1(\Gamma, A) \longleftarrow \in \longrightarrow & \text{Holonomy Class } [g] \in A^{\beta_1} \\
 & \downarrow \text{Parameterizes} \quad \uparrow \text{EM / Fisher Pullback} & \\
 \textbf{Layer 2 (Probability)} & H^1(\mathcal{S}; \mathcal{F}_\alpha) \longleftarrow \in \longrightarrow & \text{Information Curvature } I = \delta_t H \\
 & \downarrow \text{Sampling / Support} \quad \uparrow \text{Empirical Measure } \hat{P}_N & \\
 \textbf{Layer 3 (Outcomes)} & \tilde{H}^1(\mathcal{U}, \mathcal{F}_{\mathbb{Z}}) \longleftarrow \in \longrightarrow & \text{Sheaf Obstruction } \gamma
 \end{array} \tag{1}$$

1. **Layer 1 (Operation Network):** Governed by an operation system  $(\Gamma, A, g)$ , where  $\Gamma = (V, E)$  is a finite connected graph,  $A$  is an abelian coefficient group, and  $g \in Z^1(\Gamma, A)$  assigns local transformations to edges. The fundamental operational invariant is the holonomy  $h(L) = (\delta^1 g)(L) \in A^{\beta_1}$ .

2. **Layer 2 (Continuous Probability Simplex):** Governed by Baudot–Bennequin–Vigneaux (BBV) information cohomology over an information site  $\mathcal{S}$ . For a continuous probability distribution  $P_\theta \in \Lambda^{|V|-1}$ , Shannon entropy  $H$  acts as an exact 1-cocycle ( $\delta H = 0$ ), while mutual information  $I(S_0; S_1) = (\delta_t H)(S_0, S_1)$  measures the operator defect between unweighted addition  $\delta_t$  and conditional module expectation  $\delta$ .
3. **Layer 3 (Discrete Outcome Sheaf):** Governed by Abramsky–Mansfield–Barbosa (AMB) sheaf contextuality over a measurement cover  $\mathcal{U}$ . It evaluates discrete, realized support sections using integer coefficients  $\mathcal{F}_{\mathbb{Z}}$ . Non-extendability of local sections to a global joint section manifests as a non-zero sheaf cohomology class  $\gamma \in \tilde{H}^1(\mathcal{U}, \mathcal{F}_{\mathbb{Z}})$ .

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## 2 Identifiability and Cochain Indexing

To establish the precise mathematical boundary between 0-cochains  $C^0(\Gamma, A)$  (vertex potentials  $\psi$ ) and 1-cochains  $C^1(\Gamma, A)$  (edge assignments  $g$ ), we define the explicit coboundary maps:

$$\delta^0 : C^0(\Gamma, A) \longrightarrow C^1(\Gamma, A), \quad (\delta^0 \psi)(u, v) = \psi(v) - \psi(u), \quad (2)$$

$$\delta^1 : C^1(\Gamma, A) \longrightarrow C^2(\Gamma, A), \quad (\delta^1 g)(L) = \sum_{e \in L} \pm g_e. \quad (3)$$

The space of exact 1-coboundaries (flat zero-holonomy fields) is  $B^1(\Gamma, A) = \text{im}(\delta^0)$ , while the space of 1-cocycles is  $Z^1(\Gamma, A) = \ker(\delta^1)$ . The operational cohomology group is strictly:

$$H^1(\Gamma, A) = \frac{Z^1(\Gamma, A)}{B^1(\Gamma, A)} = \frac{\ker(\delta^1)}{\text{im}(\delta^0)}. \quad (4)$$

**Proposition 2.1** (Ring-Theoretic Holonomy Identifiability). *Let  $(\Gamma, A, g)$  be an operation system over an abelian group  $A$  with underlying coefficient ring  $R$ . The holonomy  $h(L) = (\delta^1 g)(L)$  of a fundamental loop  $L \in Z_1(\Gamma)$  with edge-vector  $\ell_L \in \mathbb{Z}^{|E|}$  is identifiable from a set of path observations  $P = \{p_i\}$  if and only if*

$$\ell_L \in \text{span}_R(P). \quad (5)$$

*Specifically,  $R = \mathbb{Q}$  for  $A = \mathbb{R}$ ;  $R = \mathbb{Z}$  for  $A = \mathbb{R}/\mathbb{Z}$ ; and  $R = \mathbb{Z}/n\mathbb{Z}$  for  $A = \mathbb{Z}/n\mathbb{Z}$ .*

*Proof.* If  $\ell_L \notin \text{span}_R(P)$ , there exist two distinct operation assignments  $g^{(1)}, g^{(2)} \in Z^1(\Gamma, A)$  such that  $g^{(1)}(p_i) = g^{(2)}(p_i)$  for all  $p_i \in P$ , yet  $h^{(1)}(L) \neq h^{(2)}(L)$ . Observing each edge along a loop  $L$  traversed twice yields  $\ell_{2L} = 2\ell_L$ . Over  $R = \mathbb{Q}$  or  $R = \mathbb{Z}/3\mathbb{Z}$ , 2 is invertible, making  $\ell_L \in \text{span}_R(P)$ . Over  $R = \mathbb{Z}$  or  $R = \mathbb{Z}/4\mathbb{Z}$ , 2 is non-invertible, rendering  $h(L)$  unidentifiable due to phase-wrapping ambiguity and 2-torsion.  $\square$

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## 3 Forward and Backward Dynamical Closed-Loop Feedback

The closed loop operates via a forward generative pushforward and a backward empirical pull-back:

### 3.1 Forward Pass: Operations $\longrightarrow$ Probability $\longrightarrow$ Outcomes

1. **Information Shift:** Updating an underlying parameter  $\theta \in \mathcal{M}_\theta$  shifts the operation cocycle  $g_\theta \in Z^1(\Gamma, A)$ .
2. **Probability Conditioning:** The updated operational state induces a continuous distribution  $P_\theta$  on the probability simplex  $\Lambda^{|V|-1}$  via the module action:

$$(S_0 \cdot F)(P_\theta) = \sum_j P_\theta(S_0 = v_j) F(P_\theta \mid S_0 = v_j). \quad (6)$$

This alters the continuous information curvature  $I(\theta) = \delta_t H(P_\theta)$ .

3. **Outcome Sampling:** Realizing discrete outcomes  $x \sim P_\theta$  defines the support sets  $\text{supp}(P_\theta)$ . Crossing support boundaries changes local sections  $\sigma \in \mathcal{F}_\mathbb{Z}$ , inducing topological transitions in the sheaf contextuality class  $\gamma(\theta) \in \tilde{H}^1$ .

### 3.2 Backward Pass: Outcomes $\longrightarrow$ Probability $\longrightarrow$ Operations

1. **Empirical Pullback:** Realized discrete outcome observations  $\{x_1, \dots, x_N\}$  form an empirical measure on the simplex:

$$\hat{P}_N = \frac{1}{N} \sum_{k=1}^N \delta_{x_k}. \quad (7)$$

2. **Information Geometry Step:** The empirical measure  $\hat{P}_N$  drives a natural gradient step along the Fisher-Rao metric  $g_{ij}(\theta)$  or Bregman divergence landscape:

$$\theta^{(t+1)} = \theta^{(t)} - \eta g^{ij}(\theta) \nabla_j D_{\text{KL}} \left( \hat{P}_N \parallel P_{\theta^{(t)}} \right). \quad (8)$$

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## 4 Expectation-Maximization (EM) Model Class Auditing

To audit whether an Expectation-Maximization (EM) optimization process preserves or erases operational holonomy, we evaluate optimization over a parameterized latent transition space  $\mathcal{M}_\theta$ . Let  $X$  denote observed context outcomes and  $Z$  denote latent operational states across edges  $e \in E(\Gamma)$ :

$$\textbf{E-Step: } Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z \mid X, \theta^{(t)}} [\log P(X, Z \mid \theta)] = \sum_{e \in E} \sum_z \hat{P}_N(X_e = z) \log P(g_\theta(e) = z), \quad (9)$$

$$\textbf{M-Step: } \theta^{(t+1)} = \arg \max_{\theta \in \mathcal{M}_\theta} Q(\theta \mid \theta^{(t)}). \quad (10)$$

### 4.1 The Preservation vs. Erasure Dichotomy

The structural geometry of the latent parameter manifold  $\mathcal{M}_\theta$  dictates the fate of the operational class  $[g] \in H^1(\Gamma, A)$ :

**Theorem 4.1** (Latent Holonomy Erasure under Flat Model Classes). *Let  $\mathcal{M}_\theta \subseteq \text{im}(\delta^0) = B^1(\Gamma, A)$  be a globally integrable (flat) model class, where candidate edge assignments are derived from scalar vertex potentials  $g_\theta = \delta^0 \psi_\theta$ . Then for any empirical dataset  $\hat{P}_N$ , the converged EM estimate  $\hat{\theta} = \lim_{t \rightarrow \infty} \theta^{(t)}$  forces the estimated holonomy to vanish identically:*

$$\hat{h}(L) = (\delta^1 g_{\hat{\theta}})(L) = (\delta^1 \delta^0 \psi_{\hat{\theta}})(L) \equiv 0 \quad \forall L \in Z_1(\Gamma). \quad (11)$$

*Proof.* Because candidate assignments are restricted to  $\mathcal{M}_\theta \subseteq \text{im}(\delta^0)$ , every candidate 1-cochain  $g_\theta$  is an exact 1-coboundary  $\delta^0\psi_\theta$ . Since the coboundary operator satisfies  $\delta^1 \circ \delta^0 = 0$  identically, evaluating  $g_{\hat{\theta}}$  around any fundamental cycle  $L$  yields  $(\delta^1\delta^0\psi_{\hat{\theta}})(L) = 0$ . The M-step acts as an orthogonal projection onto  $B^1(\Gamma, A)$ , recovering the exact optimization-theoretic realization of single-source global section erasure.  $\square$

**Theorem 4.2** (Holonomy Preservation under Contextual Model Classes). *Let  $\mathcal{M}_\theta \subseteq \ker(\delta^1) = Z^1(\Gamma, A)$  be an unconstrained 1-cocycle model class such that  $\mathcal{M}_\theta \not\subseteq \text{im}(\delta^0)$ . Then EM optimization converges to a maximum-likelihood estimate  $\hat{g}_{\hat{\theta}} \in Z^1(\Gamma, A)$  that preserves non-trivial operational holonomy  $[\hat{g}] \neq 0 \in H^1(\Gamma, A)$ .*

*Proof.* Since edge assignments are parameterized independently without forcing single-source vertex potential integrability  $g \in \text{im}(\delta^0)$ , the M-step maximizes likelihood across  $Z^1(\Gamma, A)$ . When empirical path discrepancies reflect non-zero holonomy,  $\hat{g}_{\hat{\theta}} \notin \text{im}(\delta^0)$ , yielding  $[\hat{g}] \neq 0$ .  $\square$

## 5 Statistical Auditing & Exponential Sample Complexity

To distinguish a true non-zero holonomy class  $[\hat{g}] \neq 0$  from estimation noise under continuous dispersion  $\sigma$ , we employ the calibrated circular  $\chi^2$  statistic.

Using the circular variance  $S = 1 - |\mathbb{E}[e^{i\theta}]|$ , the  $\chi^2$  test statistic is calibrated at 4.2–6.3% false-positive rate under  $\alpha = 0.05$  across directional noise levels  $\sigma \in [0.05, 0.35]$  turns.

**Proposition 5.1** (Sample-Size Complexity Threshold). *The number of observations per edge  $N_{80}$  required to achieve 80% statistical power in detecting a loop holonomy  $h(L) \neq 0$  scales as:*

$$N_{80} \propto |L| \cdot \exp(4\pi^2\sigma^2), \quad (12)$$

where  $|L|$  is the fundamental loop length and  $\sigma^2$  is the variance of the edge-level noise.

The exponential factor  $\exp(4\pi^2\sigma^2)$  reflects the information-geometric phase transition where circular wrapping around  $S^1$  exponentially degrades the Fisher information available for operational loop recovery.

## 6 Summary of Integrated Mathematical Structures

Table 1: Summary of Stratified Layers and Associated Mathematical Invariants

| Layer / Domain       | Algebraic Domain                        | Coboundary / Operator                             | Fundamental Invariant                                                 |
|----------------------|-----------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| Layer 1: Operations  | $C^1(\Gamma, A)$                        | $\delta^1 : C^1 \rightarrow C^2$                  | $[g] \in H^1(\Gamma, A) = \frac{\ker(\delta^1)}{\text{im}(\delta^0)}$ |
| Layer 2: Probability | $\mathcal{S}_k(\Lambda^n)$              | $d : \mathcal{S}_k \rightarrow \mathcal{S}_{k+1}$ | $I = \delta_t H - dH$                                                 |
| Layer 3: Outcomes    | $\mathcal{F}_{\mathbb{Z}}(\mathcal{U})$ | $\delta^0 : C^0 \rightarrow C^1$                  | $\gamma \in \check{H}^1(\mathcal{U}, \mathcal{F}_{\mathbb{Z}})$       |